

UC BERKELEY CS 180 · FALL 2026 · PROJECT 2

Fun with Filters and Frequencies

From a four-loop convolution to frequency-selective hybrids and multiresolution splines: a reproducible study of how local filters become visual structure.

[Parts 1.1–2.4 complete](#)[3 hybrid images](#)[3 multiresolution blends](#)[Bells & Whistles](#)dasheng-lab.github.io/pj2/**10⁻¹⁵**

convolution agreement with SciPy

6

Gaussian/Laplacian levels, no downsampling

3 + 3

hybrid images and spline blends

2

Bells & Whistles: orientation + color

PART 1 • FUN WITH FILTERS

1.1 Convolutions from Scratch

10 pts

I implemented discrete 2D convolution twice using only NumPy: a literal four-loop version and a two-loop version whose inner kernel product is vectorized. Both explicitly zero-pad to preserve the input size and flip the kernel in both axes, so they implement convolution rather than cross-correlation.

```
def convolve_four_loops(image, kernel):
    padded = np.pad(image, padding)
    flipped = kernel[::-1, ::-1]
    for y in range(H):
        for x in range(W):
            for j in range(kh):
                for i in range(kw):
                    out[y,x] += (
                        padded[y+j,x+i] * flipped[j,i])
    return out
```

```
def convolve_two_loops(image, kernel):
    padded = np.pad(image, padding)
    flipped = kernel[::-1, ::-1]
    for y in range(H):
        for x in range(W):
            patch = padded[y:y+kh, x:x+kw]
            out[y,x] = np.sum(patch * flipped)
    return out
```

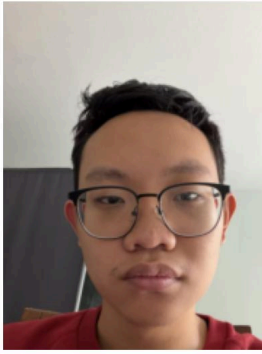
Correctness and timing

Method, 96×96 and 9×9	Time	max error
Four loops	0.646 s	8.88×10^{-16}
Two loops	0.098 s	4.44×10^{-16}
SciPy convolve2d	0.0024 s	reference

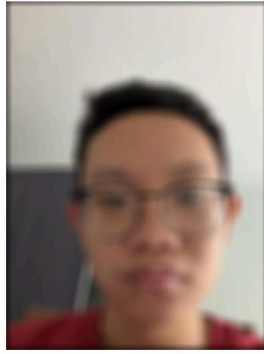
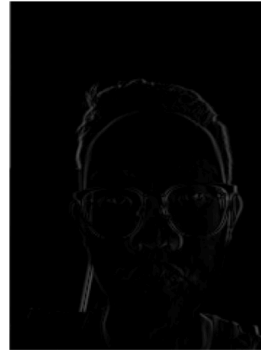
Interpretation

Vectorizing the 81 multiply-adds gives a $6.6\times$ improvement over four Python kernel loops. SciPy remains much faster because its loop nest is compiled. The floating-point discrepancies are at machine precision, including the RGB comparison (6.66×10^{-16}).

Original selfie



9×9 box blur (our convolution)

 $|D_x * I|$  $|D_y * I|$ 

A 9×9 box filter removes fine facial texture; horizontal and vertical finite differences isolate orthogonal transitions.

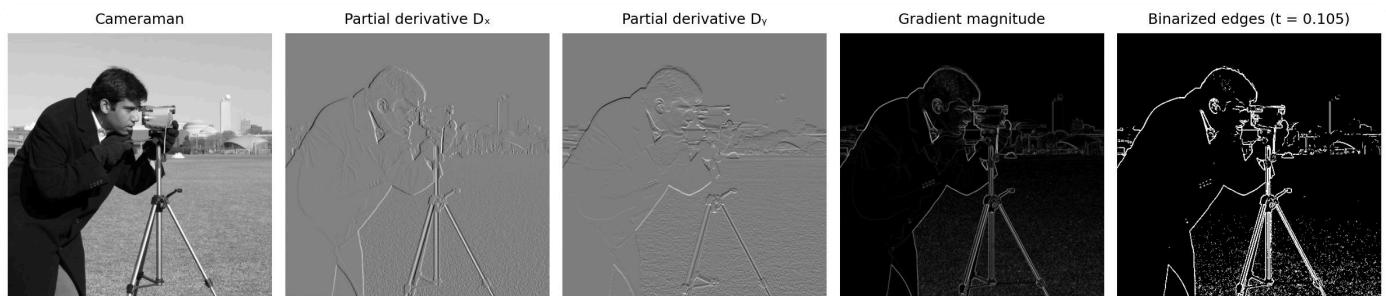
PART 1 • DERIVATIVES

1.2 Finite Difference Operator

10 pts

I convolved cameraman with centered finite differences $D_x = [1, 0, -1]/2$ and $D_y = D_x^T$, then formed the gradient magnitude $\sqrt{I_x^2 + I_y^2}$. A threshold of **0.105** retains the camera, tripod, coat, and silhouette while suppressing weak grass and film noise; 5.83% of pixels remain.

$$m(x,y) = \sqrt{I_x(x,y)^2 + I_y(x,y)^2} \quad \cdot \quad \text{edge}(x,y) = [m(x,y) > 0.105]$$



D_x , D_y , gradient magnitude, and the selected binary edge map. The signed derivative panels use mid-gray for zero.

PART 1 • SMOOTHING BEFORE DIFFERENTIATION

1.3 Derivative of Gaussian

10 pts

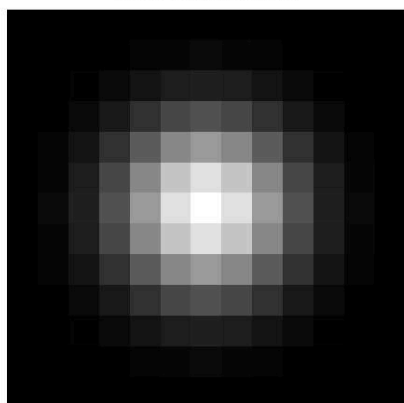
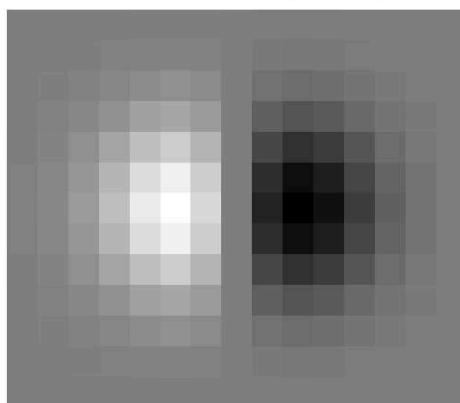
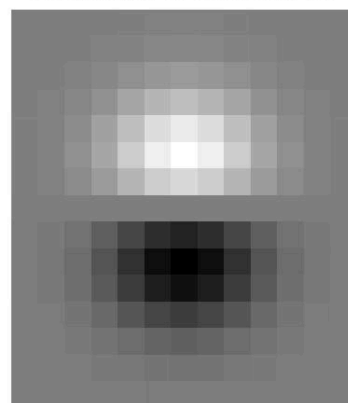
A 13×13 Gaussian with $\sigma=2$ suppresses high-frequency noise before differentiation. The resulting edges are thicker and more continuous than raw finite differences. By associativity, differentiating the Gaussian once creates two derivative-of-Gaussian filters that can be convolved with the image directly.

Image I

Gaussian G_σ  $D_{x/y}$

Numerical verification. Blur-then-differentiate and one-pass DoG agree to 4.64×10^{-16} in x and 5.13×10^{-16} in y. Thus the visible outputs are the same up to rounding.

Gaussian kernel

Derivative-of-Gaussian D_x Derivative-of-Gaussian D_y 

The Gaussian and its x/y derivative filters. Opposite signs appear on opposite sides of each DoG.

Original

Gaussian blur ($\sigma=2$)Blur $\rightarrow$ gradient magnitudeSmoothed edges ($t=0.050$)DoG $\rightarrow$ gradient magnitudeDoG edges ($t=0.050$)

Two-pass smoothing/derivative and one-pass DoG produce equivalent magnitudes and binary edges at $t=0.050$.

BELLS & WHISTLES

Gradient Orientation without Angle APIs

Extra

I approximated atan algebraically on $|z| \leq 1$ using $z(\pi/4 + 0.273(1 - |z|))$, used the reciprocal identity outside that interval, and handled quadrants explicitly from the signs of I_x and I_y . This avoids atan and atan2 . Orientation maps to HSV hue, while clipped gradient magnitude controls value, so flat regions remain dark and strong edges reveal direction.

Cameraman



Gradient magnitude



Orientation (hue), strength (value)



Hue cycles through edge direction; brightness encodes edge strength. Similar colors along the subject outline confirm consistent local orientation.

PART 2 • FUN WITH FREQUENCIES

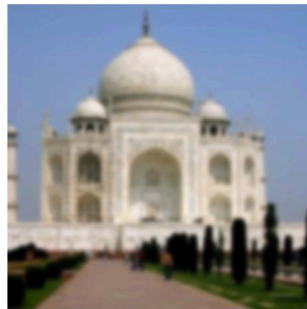
2.1 Image Sharpening

15 pts

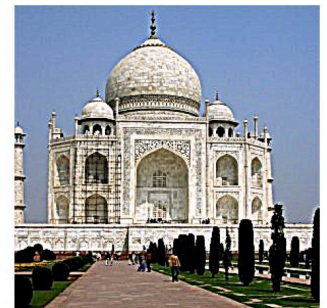
The unsharp mask treats $I - G_\sigma * I$ as high-frequency detail and adds α times that signal back. I used $\sigma=2$ and $\alpha=1.5$ for the main Taj result. Larger α makes stone edges crisper but eventually introduces bright/dark halos.

$$I_{\text{sharp}} = I + \alpha(I - G_\sigma * I) = ((1+\alpha)\delta - \alpha G_\sigma) * I$$

Original Taj

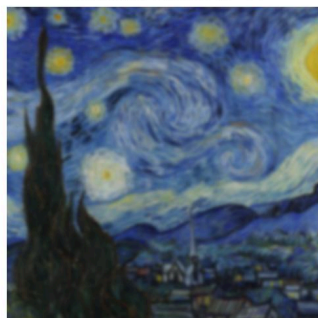
Low frequencies $G_\sigma * I$ High frequencies $I - G_\sigma * I$ Sharpened $I + 1.5 \cdot \text{high}$ 

Original, low-frequency component, signed high-frequency component, and the final sharpened image.

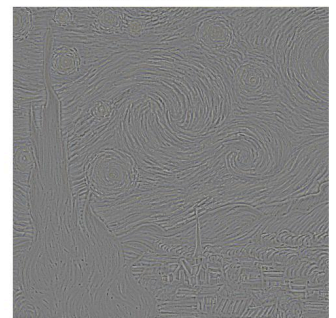
 $\alpha = 0.5$  $\alpha = 1.0$  $\alpha = 1.5$  $\alpha = 2.5$ 

Increasing α from 0.5 to 2.5. $\alpha=1.5$ balances detail and ringing.

Originally sharp

Artificially blurred ($\sigma=3$)Re-sharpened ($\alpha=2$)

Residual vs. original (+0.5)



A sharp Starry Night was blurred and re-sharpened. PSNR improves from 22.64 dB to 23.68 dB, but the residual shows that deleted frequencies cannot be exactly recovered.

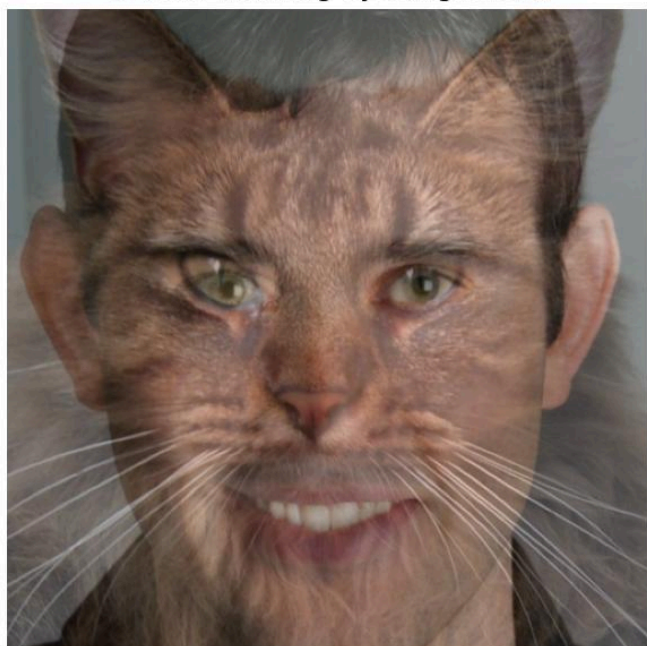
PART 2 • SCALE-DEPENDENT PERCEPTION

2.2 Hybrid Images

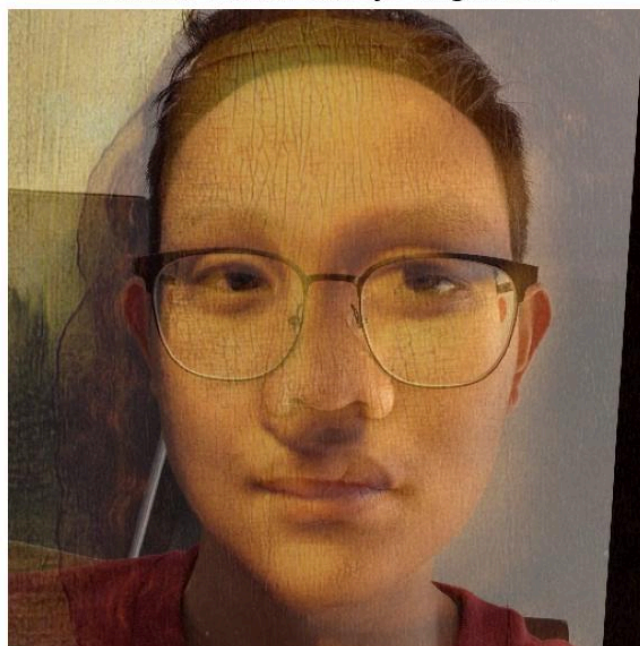
20 pts

Each pair is geometrically aligned before filtering. For both face pairs, I manually verified the two eye centers and mapped them to the same target coordinates with a similarity transform, which corrects translation, scale, and in-plane rotation without distorting facial proportions. One image contributes a Gaussian low-pass and the other contributes $I - G_\sigma * I$; close viewing resolves high-frequency edges, while distance reveals the low-frequency subject.

Derek / Nutmeg eye alignment

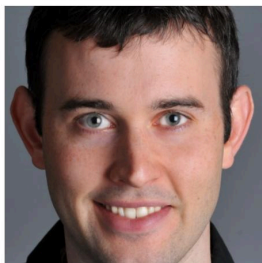


Selfie / Mona Lisa eye alignment

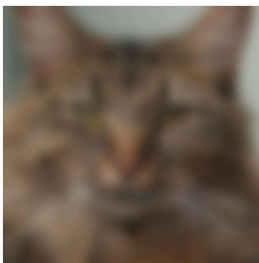
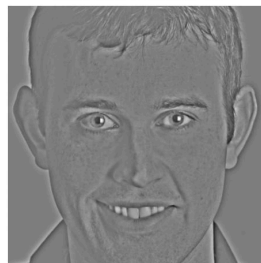


Alignment verification. These unmarked 50/50 overlays show that both pupils coincide after the similarity transform. Every original, filtered, Fourier, and final hybrid panel below is generated from these same aligned arrays.

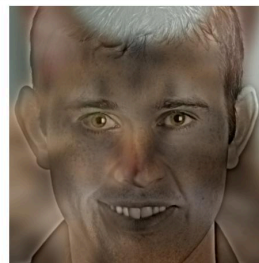
Derek aligned (high-pass)



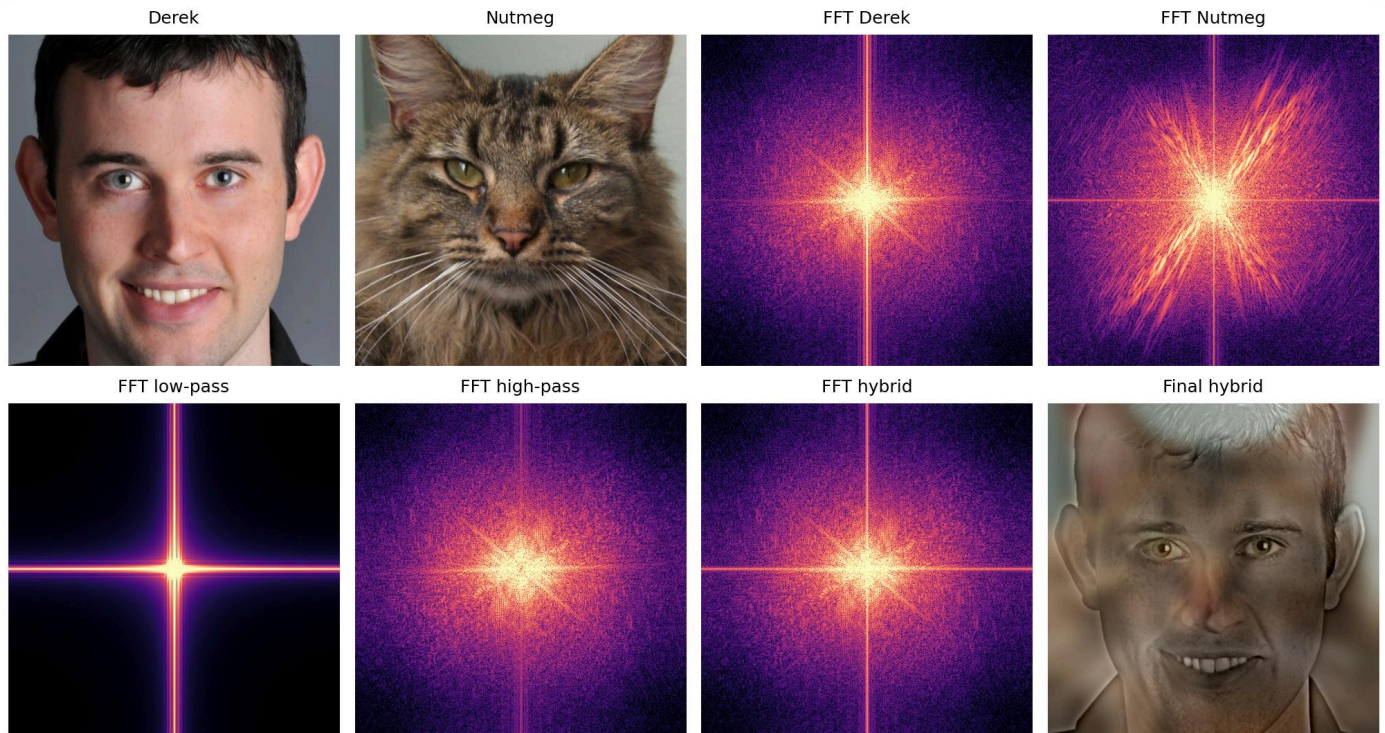
Nutmeg aligned (low-pass)

Low-pass Nutmeg ($\sigma=12$)High-pass Derek ($\sigma=6$)

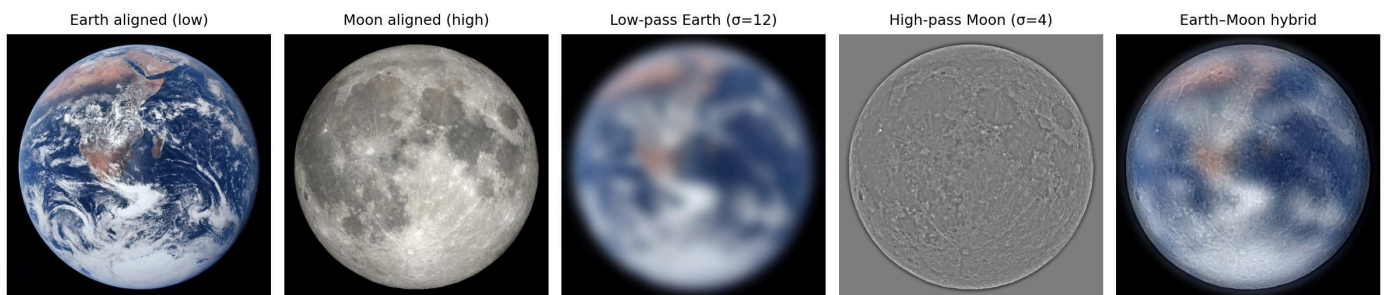
Hybrid: Derek up close, Nutmeg far away



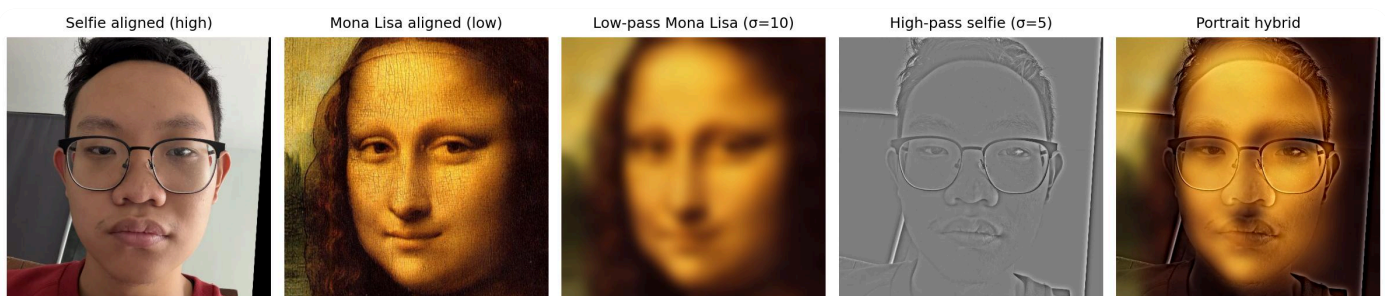
Required pair. Low-pass Nutmeg $\sigma=12$; high-pass Derek $\sigma=6$. Eye landmarks determine a reproducible similarity transform.



Favorite analysis. Log-magnitude spectra show low-pass energy concentrated at the origin, high-pass suppression at the origin, and their complementary sum in the hybrid.



Earth low-pass $\sigma=12$ + Moon high-pass $\sigma=4$. Automatic disc crops align the horizons.



Mona Lisa low-pass $\sigma=10$ + selfie high-pass $\sigma=5$. The same eye-centered similarity transform aligns both faces.

Cutoff selection. I increased the low-pass σ until small texture no longer competed at close range, then increased the high-pass σ until identity survived without a large luminance halo. These are spatial

standard deviations in pixels; larger σ corresponds to a lower frequency cutoff.

Bells & Whistles: color contribution

Color is most stable in the low-frequency component: broad chromatic fields survive downsampling and support the far-view identity. High-pass-only color is fragile and adds colored edge halos. I therefore use low-pass color with grayscale high frequencies for the displayed hybrids.



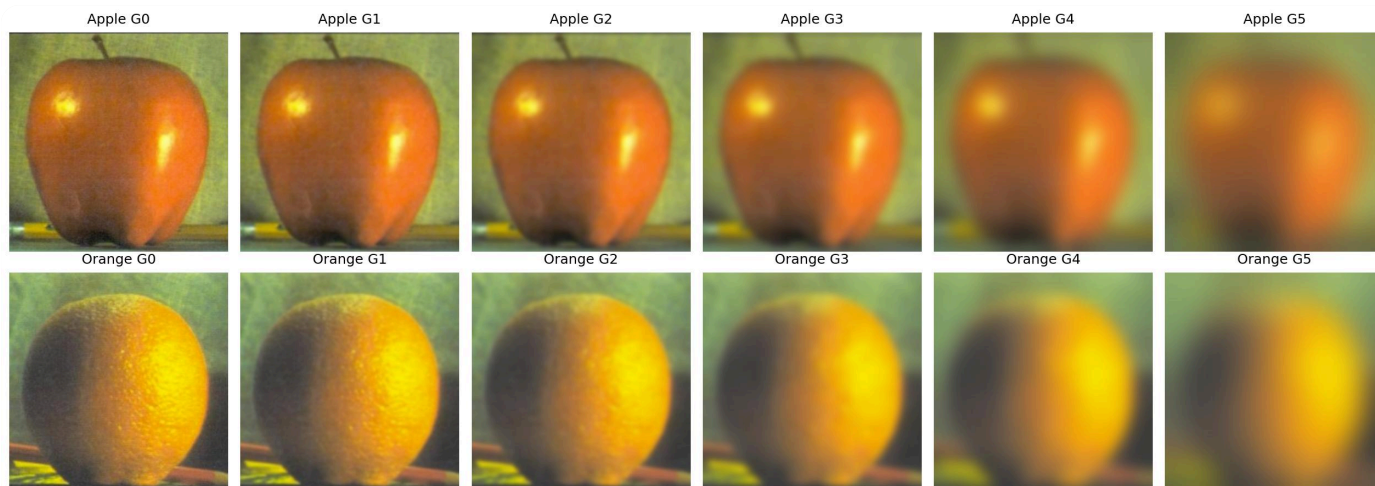
Both-color, low-only color, high-only color, and grayscale variants under identical cutoffs.

PART 2 • FREQUENCY BANDS

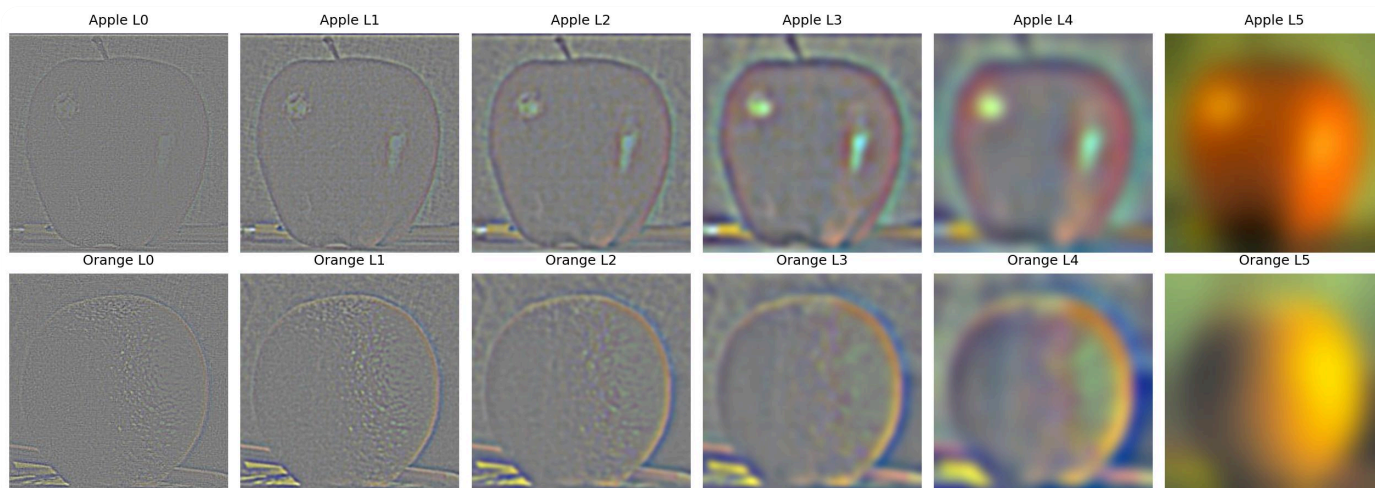
2.3 Gaussian & Laplacian Stacks

10 pts

I constructed six-level stacks without downsampling or any pyramid routine. $G_0=I$ and later levels blur the original with $\sigma=2,4,8,16,32$. Each Laplacian band is $L_i=G_i-G_{i+1}$, with the coarsest Gaussian retained as the residual. Summing all bands reconstructs both inputs with maximum error 1.11×10^{-16} .



Gaussian stacks: spatial dimensions stay fixed while fine structure progressively disappears.



Laplacian stacks: each panel isolates a frequency band; the last panel is the low-frequency residual.

PART 2 · MULTIREOLUTION SPLINE

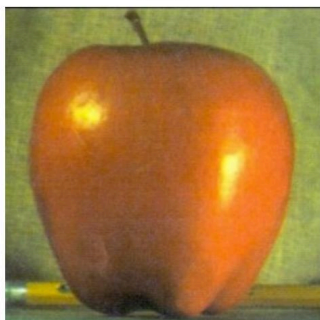
2.4 Multiresolution Blending

20 pts

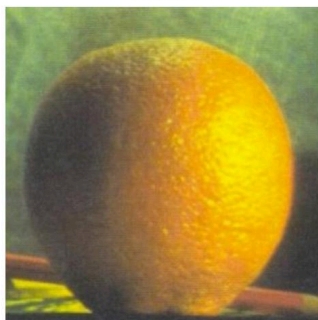
For every level, I blur the binary or irregular mask to the matching scale and combine Laplacian bands as $G(M)_i L(A)_i + (1 - G(M)_i) L(B)_i$. Summing the blended bands produces a seam whose width is appropriate to each spatial frequency.

$$\text{Blend} = \sum_i [G(M)_i \cdot L(A)_i + (1 - G(M)_i) \cdot L(B)_i]$$

Apple



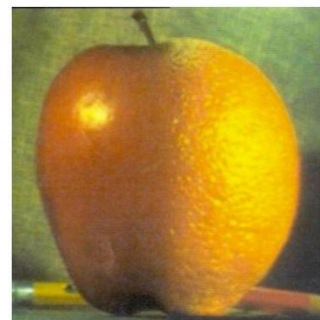
Orange



Binary seam mask



Orapple



The standard Orapple with a vertical mask. Low frequencies transition broadly while high frequencies stay localized.

Earth



Moon



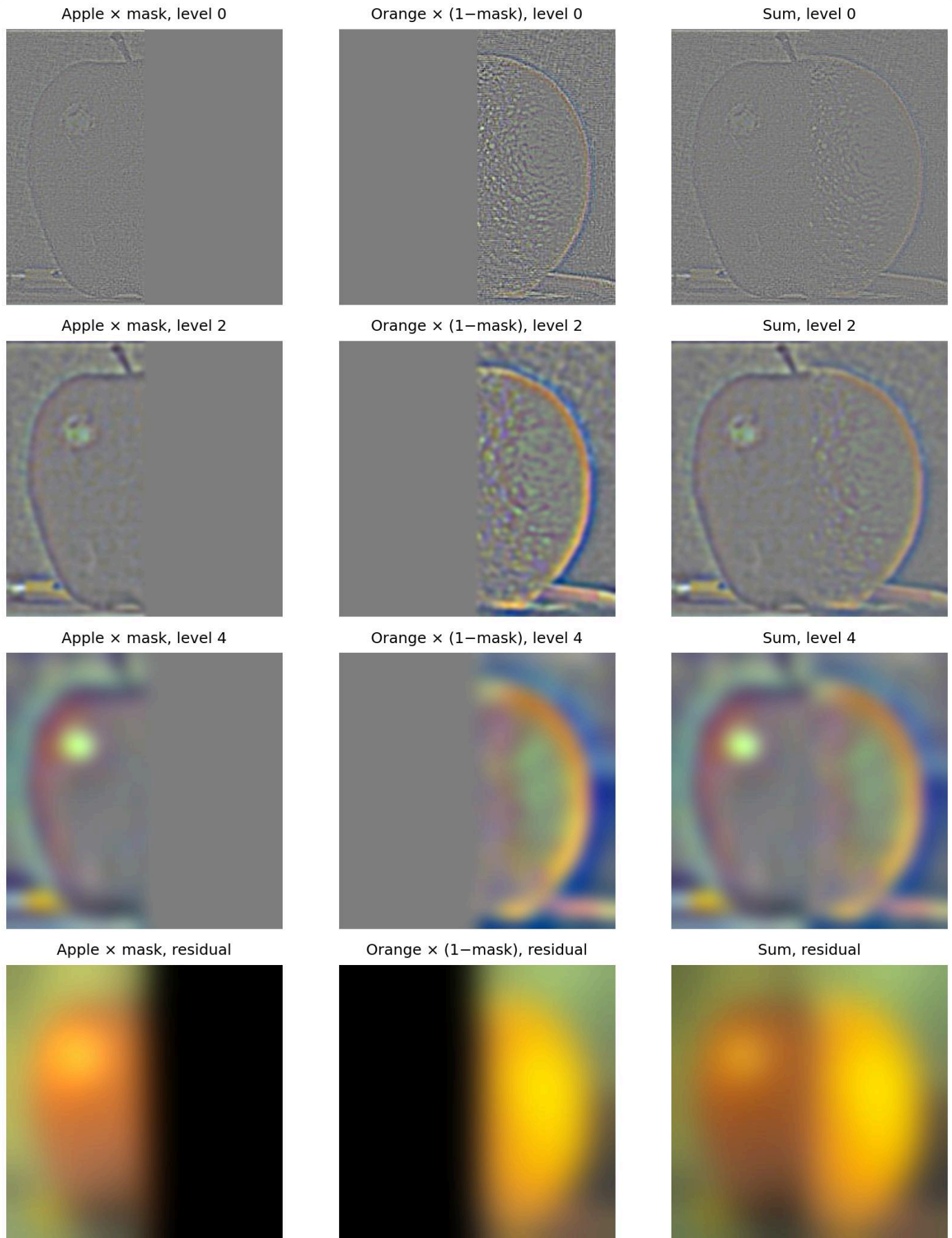
Horizontal mask



Earth/Moon multiresolution blend



Custom blend 1: a horizontal Earth/Moon composition with a smooth equatorial transition.



Recreation of the Figure 3.42 process: masked Apple band, masked Orange band, and their sum at representative high, middle, low, and residual levels.

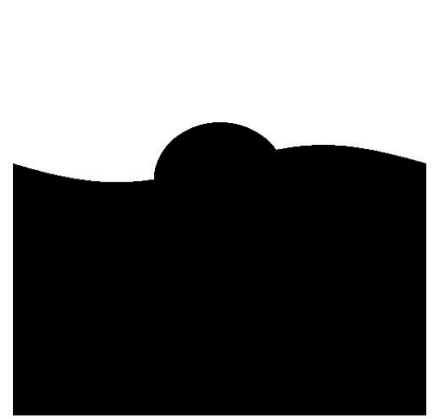
Starry Night



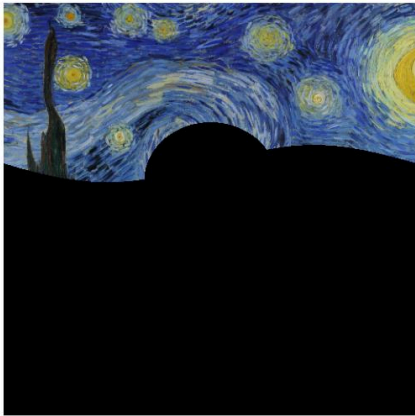
Taj Mahal



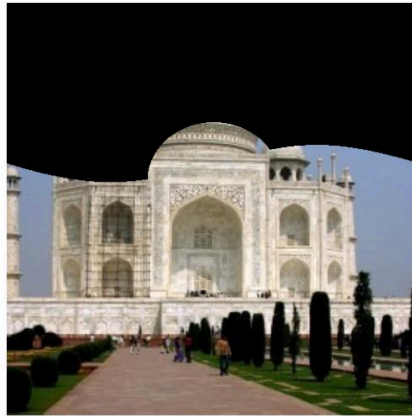
Irregular sky mask



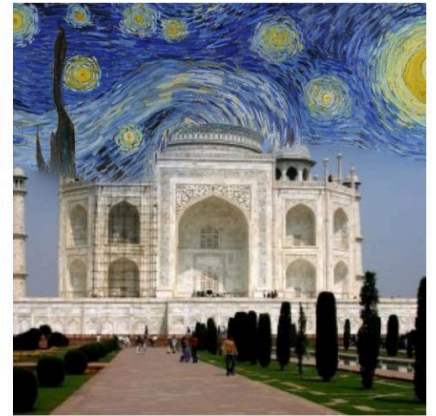
Masked stars



Masked Taj



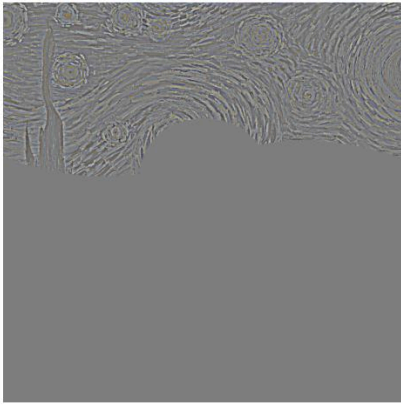
Final blend



Custom blend 2 and favorite. A sinusoidal skyline mask with an elliptical dome exclusion replaces the sky but preserves the monument silhouette—an irregular mask rather than a straight split.

Color blending. Every Gaussian and Laplacian operation is applied channel-wise in RGB. Blurring the mask—not the images after compositing—lets the same spatially varying transition act consistently on all three channels.

Stars band L0



Taj band L0



Combined L0



Stars band L2



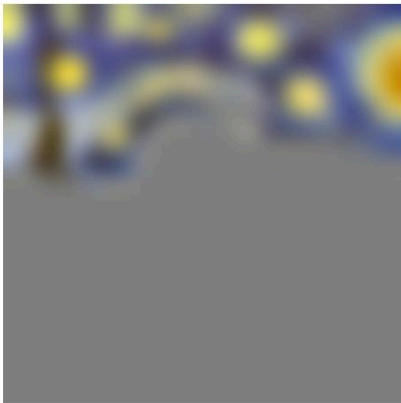
Taj band L2



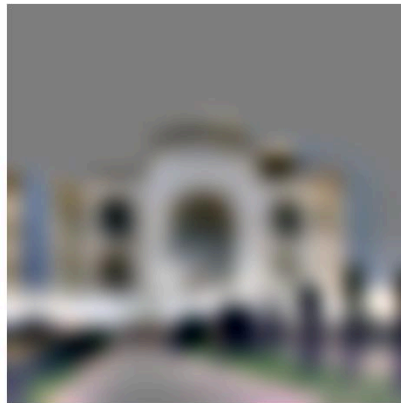
Combined L2



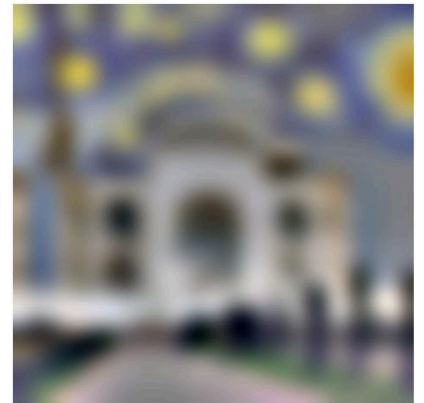
Stars band L4



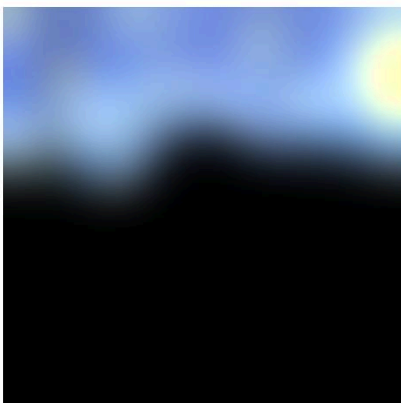
Taj band L4



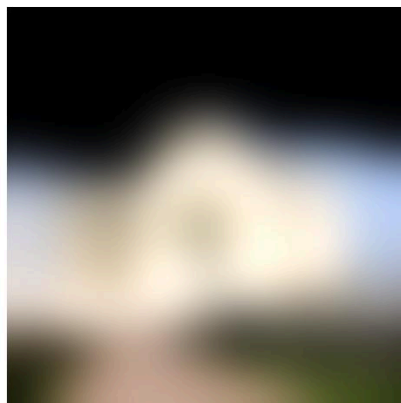
Combined L4



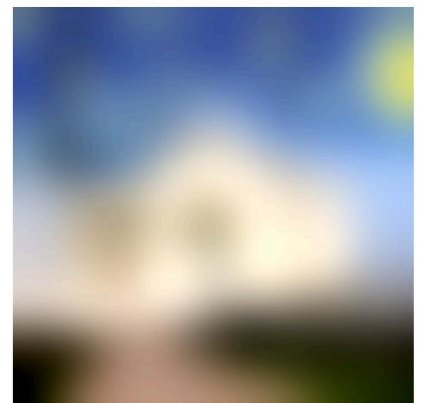
Stars band L5



Taj band L5



Combined L5



Representative Laplacian levels for the irregular blend: masked stars, masked Taj, and the combined band. Coarse levels establish illumination; fine levels preserve architecture and brush texture.

REFLECTION

Most Important Thing I Learned

Clarity

The most important insight is that “an image” is not a single object at all scales. The same Gaussian family can separate detail for sharpening, choose which identity survives viewing distance in a hybrid, and determine how wide a seam should be in each Laplacian band. Thinking in frequency bands turns several seemingly different editing tricks into one coherent operation: preserve, suppress, or spatially route selected scales.

Reproducibility and sources

The submitted code/main.py regenerates every figure and records parameters and numerical checks in results/metrics.json. Alignment points, face crops, disc crops, thresholds, and masks are deterministic; no interactive clicks or network access are required.

Course materials: [CS 180 Project 2 specification](#) and the [official spline sample](#). External imagery: NASA’s [Apollo 17 Blue Marble](#) and [Moon phase imagery](#); public-domain [Mona Lisa](#) and [The Starry Night](#). Derek/Nutmeg, selfie, Taj Mahal, and cameraman are project/provided test inputs.

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